Ain Shams University
Faculty of Engineering
Electronics & Comm. Eng. Dept.

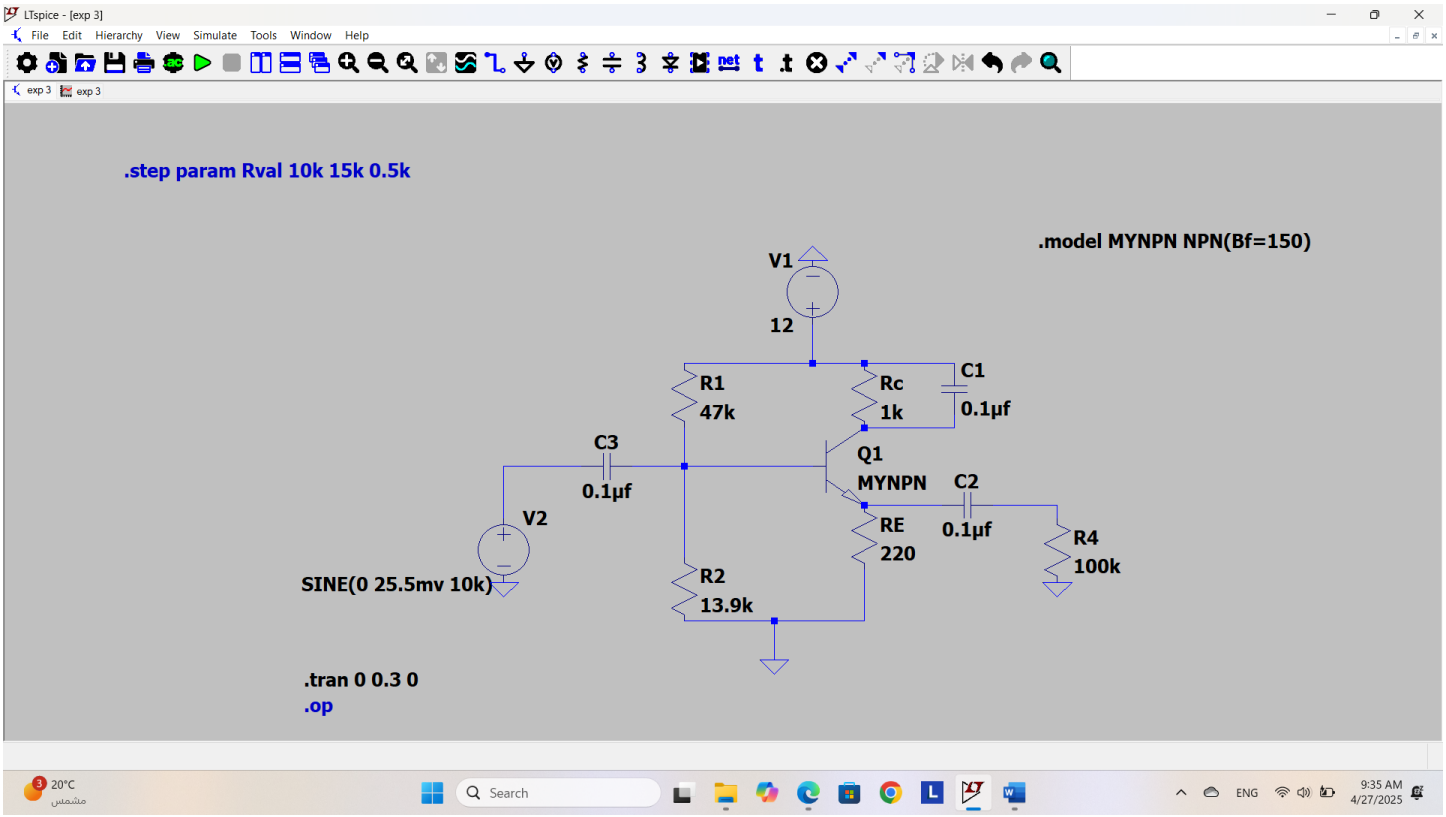


ECE211: Electronics
2nd Year Electrical Engineering
Fall 2024

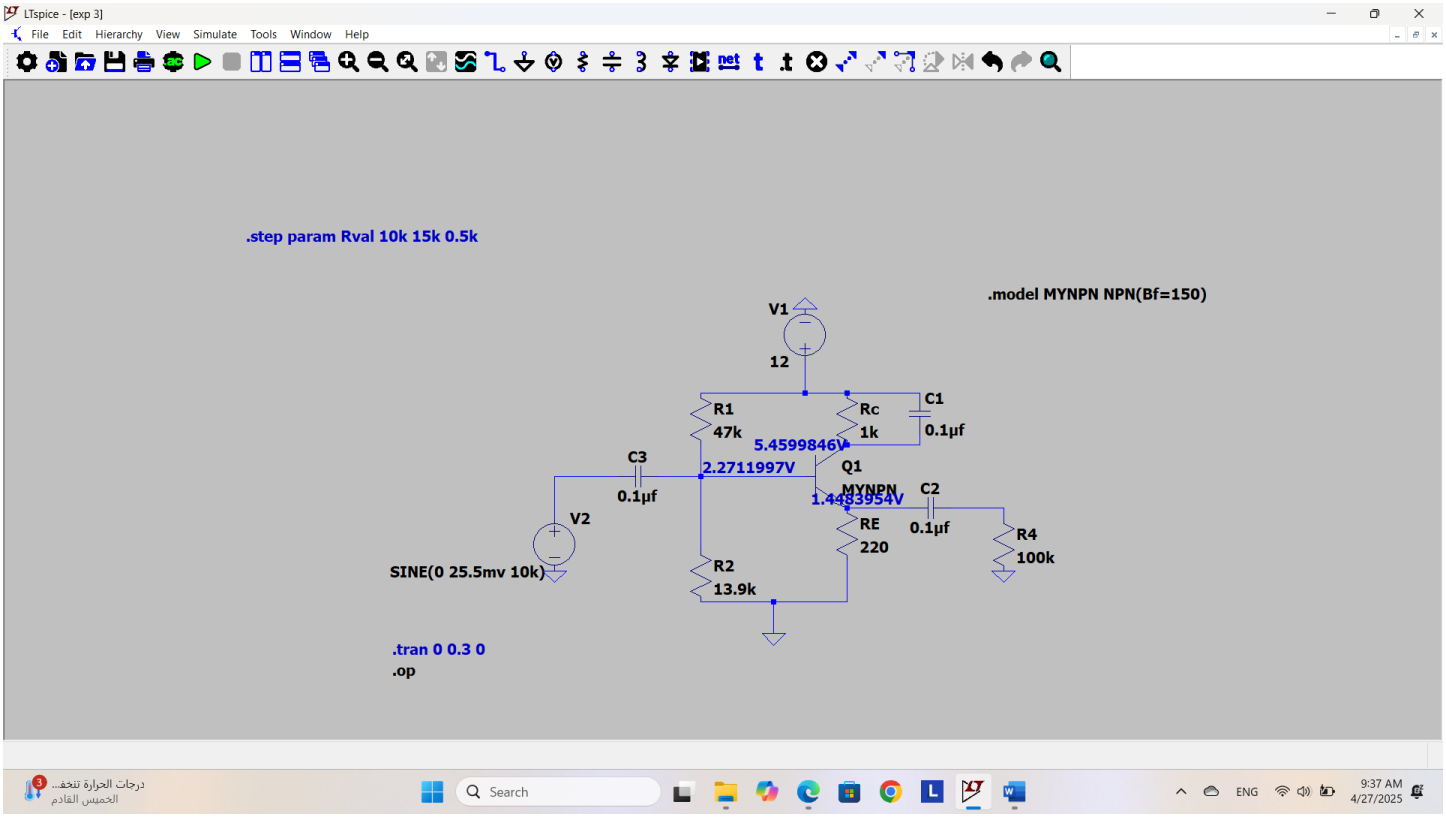
Experiment (Z) Report

Names	IDs	Department	Attendance slot	Group No.	Group Code Part A, B
Abdelrahman sherif	2300100	ZZZ	1	Z	Z,Z
Second Name	ZZZZZZZ	ZZZ			
Third Name	ZZZZZZZ	ZZZ			

Part A:


LTspice - [exp 3]
File Edit Hierarchy View Simulate Tools Window Help
exp 3
.step param Rval 10k 15k 0.5k
.model MYNPN NPN(Bf=150)
V1
12
R1
47k
Rc
1k
C1
0.1µf
C3
0.1µf
Q1
MYNPN
C2
0.1µf
V2
RE
220
R4
100k
SINE(0 25.5mv 10k)
R2
13.9k
.tran 0 0.3 0
.op
Search
ENG
9:35 AM
4/27/2025

Part B:

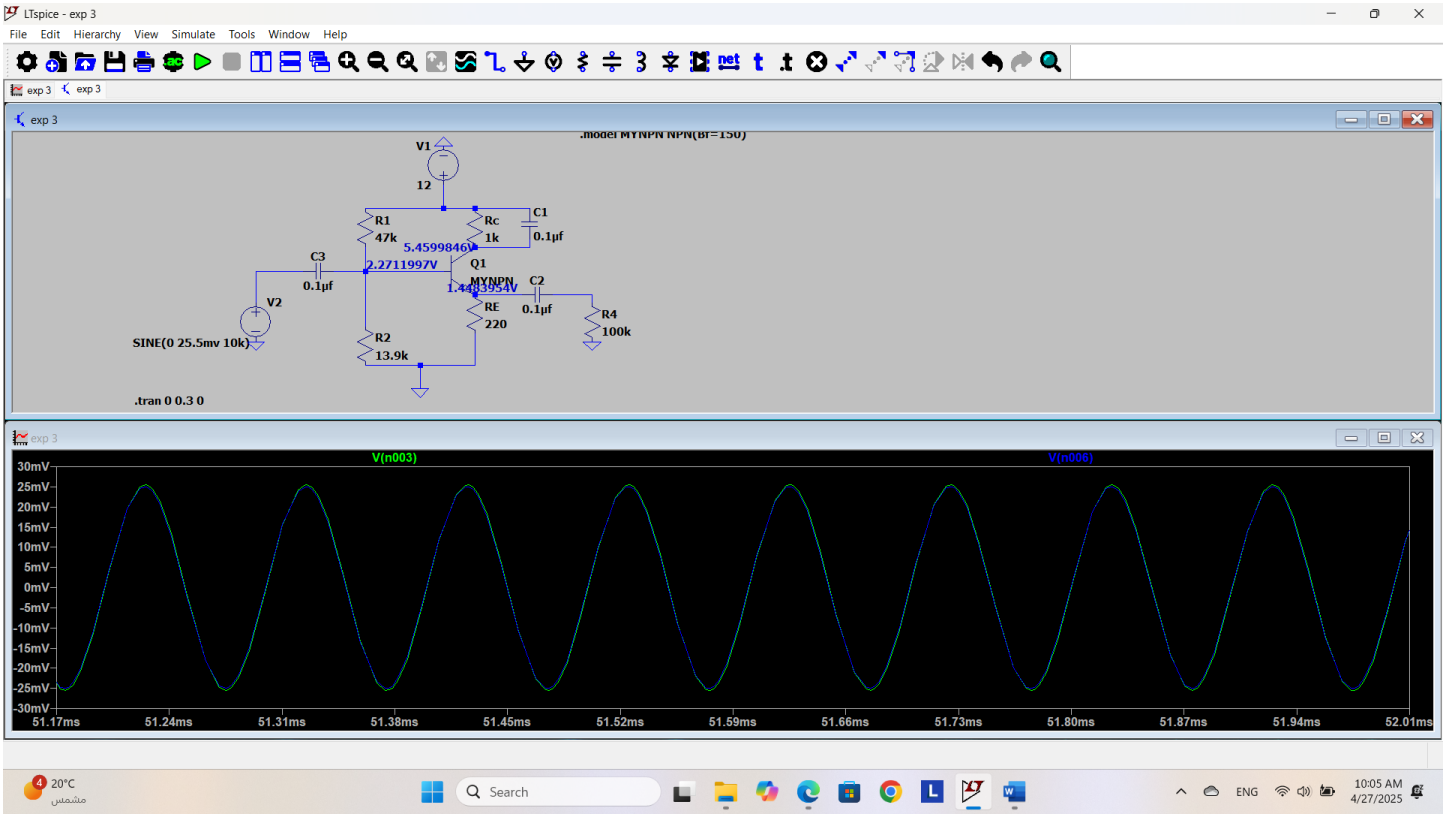

LTspice - [exp 3]
File Edit Hierarchy View Simulate Tools Window Help
.step param Rval 10k 15k 0.5k
.model MYNPN NPN(Bf=150)
V1
12
R1
47k
Rc
1k
C1
0.1µf
5.4599846V
C3
0.1µf
2.2711997V
Q1
MYNPN
1.4483954V
C2
0.1µf
V2
RE
220
R4
100k
SINE(0 25.5mv 10k)
R2
13.9k
.tran 0 0.3 0
.op
Search
ENG
9:37 AM
4/27/2025

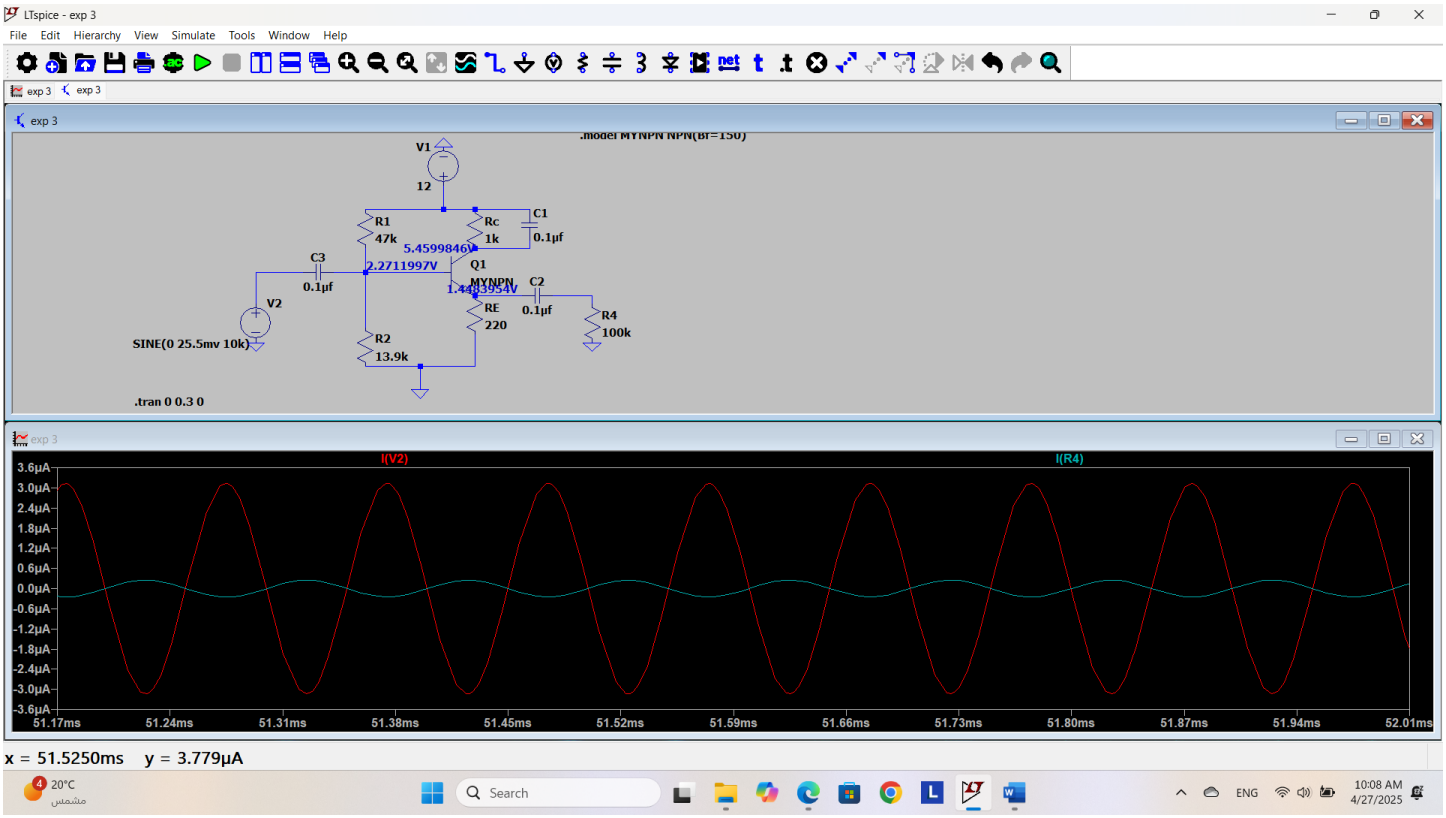
Part C:

Does cc has an effect on output waveform ?

Yes as it acts as a degenerate resistance cancelin the resistance in ac analysis by acting as short circuit while leaving it in dc analysis to stable the opersting point

voltage gain=1




LTspice - exp 3
File Edit Hierarchy View Simulate Tools Window Help
exp 3
.model MYNPN NPN(Bf=150)
V1
12
R1
47k
Rc
1k
C1
0.1µf
5.4599846V
2.2711997V
Q1
MYNPN
1.4483954V
C3
0.1µf
C2
0.1µf
V2
SINE(0 25.5mv 10k)
RE
220
R4
100k
R2
13.9k
.tran 0 0.3 0
I(V2)
I(R4)
3.6µA
3.0µA
2.4µA
1.8µA
1.2µA
0.6µA
0.0µA
-0.6µA
-1.2µA
-1.8µA
-2.4µA
-3.0µA
-3.6µA
51.17ms
51.24ms
51.31ms
51.38ms
51.45ms
51.52ms
51.59ms
51.66ms
51.73ms
51.80ms
51.87ms
51.94ms
52.01ms
x = 51.5250ms y = 3.779µA
20°C
Search
ENG
10:08 AM
4/27/2025

Part D:

$-4 \Rightarrow 4$

6.54×10^{-3}

$x[k] = e^{-j2\pi k}$ $[-4, 4]$

$x[k] = k \quad 0 < k < 4$

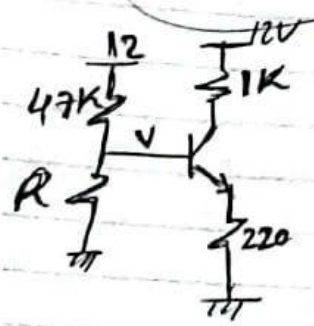
$-k \quad -4 < k < 0$

$\frac{-8\pi - 0}{4 - 0} = \frac{-8\pi}{4}$

$\alpha = \frac{\beta}{\beta+1} = \frac{150}{151}$ $\theta = \frac{-8\pi}{4} k = -2\pi k$

$\alpha < 1$ 3.524×10^{-6}

$V_{cc} = 4V$



$\beta = 150$ $\frac{I_c}{\alpha} \approx I_E$

$V - 0.7$

$12 = I_c [\beta 220] + 4V$

$12 = I_c 10^3$ $I_c = 6.56 \times 10^{-3}$

$+ I_c \frac{151}{150} \times 220$ 6.56mA

$+ 4V$ $I_c = 6.55mA$

$V_{in} = 0.7 + 6.55 \times \frac{151}{150} \times 220$

$12 \times \frac{R}{47+R} = 2.27$ $= 2.15$

$R = 10.26$ 2.27 43.5kA

$\frac{240}{43} R = 47 + R$ $\frac{12}{2.27} R - R = 47$

7.19 1.06 6.82 1.14 $V_{cc} = 12 - I_c \times (10^3 + 22c)$

5.68

1.88 1.06

I_c 5.176 $I_E = 5.21 c974$ $\cos\alpha$

$I_E \alpha = I_c$

I_C $I_B = 3.45 \times 10^{-5}$ $A_i = \frac{R_{in}}{R_L} = \frac{8.1134K}{100K}$

$I_C = 0.00$

5.17×10^{-3}

$r_\pi = \frac{\beta}{g_m}$ $g_m = \frac{I_{cq}}{V_T} = 0.252$ $R_{6_{eff}} = 219.5$

$A_i = 0.081$

$r_\pi = 596$ $R_B = 10.727K$

$R_{in} = R_B // 33301.5 = 8.1134K$

$33.3K$